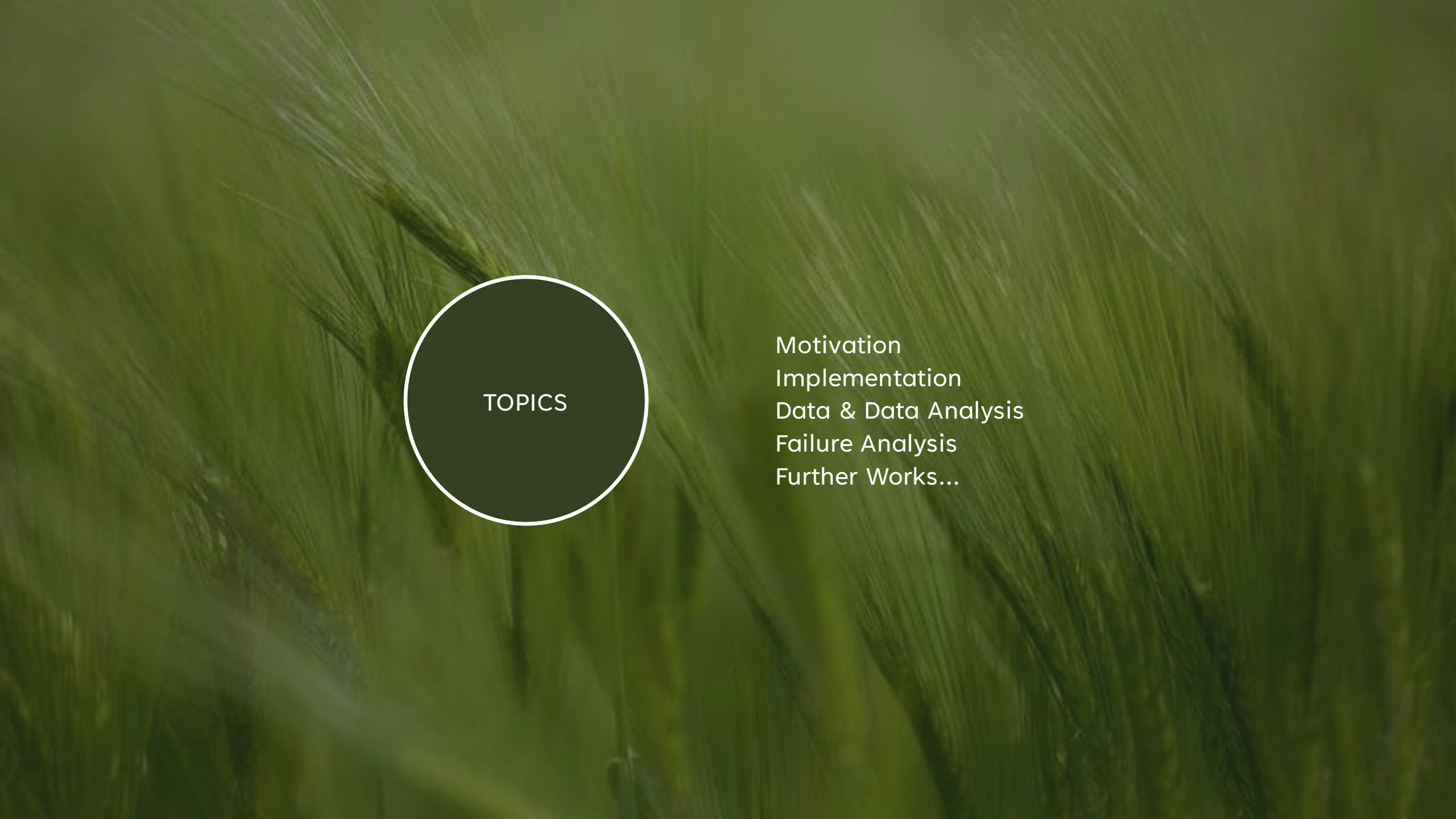

Soil Monitoring
System


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Motivation

Motivation

Due to evaporation or overwatering, traditional irrigation methods can waste significant amounts of water. This project aims to conserve water and improve plant growth by using a drip irrigation system with automated controls.

Implementation

Implementation

- Sensor Network: We implemented a sensor network to monitor soil moisture and temperature. This data is crucial for determining the plant's watering needs.
- Intelligent Control System: The system compares sensor data with the plant's specific moisture requirements. A motor is activated based on this comparison to deliver water through drip irrigation tubing directly to the plant's roots.
- Water Management:
  - A reservoir stores water for the system.
  - A water level sensor monitors the reservoir's depth, triggering a refill when necessary.
  - To prevent water waste, excess water from the drip lines is recycled back to the reservoir using a closed-loop design.

Data & Data Analysis

Data & Data Analysis

Data collected from the sensors (Moisture, Temperature, Water Level) can be used to:

- Fine-tune watering schedules for optimal plant growth.
- Identify potential issues like extreme temperatures or prolonged dryness.
- Keeping an eye on the remaining water in the reservoir

As data collected by the sensors is analyzed and compared with a threshold that may trigger the actuators

- Whenever the soil moisture is less than the threshold water pump will be triggered
- Whenever the water level is less than the threshold buzzer will be triggered

(There will be a time interval between two corresponding triggers of actuators)

Failure Analysis

Failure Analysis

- Sensor malfunction: Faulty sensors could provide inaccurate data, leading to under or overwatering.
- Clogged drip lines: Blockages in the tubing can prevent water from reaching the plant.
- Pump failure: A malfunctioning pump would disrupt the water delivery system.
- Corrosion of traces on the water resistance sensor will give inaccurate sensor readings

Setup may fail in some rare conditions :

- When soil moisture is high, but the temperature is also high

Further Works...

- Mobile App Integration: Develop a mobile app to monitor sensor data, control watering schedules, and receive alerts remotely.
- Advanced Automation: Explore integrating weather data to adjust watering based on forecasts for improved efficiency.
- System Scalability: Design the system for scalability to accommodate multiple plants or a larger garden area. Here we can also use new sensors like thermal scanners for very large fields from arial.